

Course Title	Embedded Hardware Design		
Course Code	EL203	Credit Structure	3-0-2-4 (L-T-P-C)
Category	Core	Semester	Winter Semester (AY 25-26)
Program	B.Tech. (ICT, CS, EnV)		
Prerequisites	Digital Logic and Computer Organization		
Course Objectives/ Brief Course Description	This course will give the students an introductory understanding of embedded systems which includes the programming the AVR microcontroller/ARM processor based Microcontrollers, VHDL, Hardware/Software Co-Design, interfacing of external peripheral devices to microcontroller, and troubleshooting the embedded systems (PO1, PO3, PSO1). The course consists of two parts which are (i) interactive lecture section and (ii) hand-on which will provide a platform to develop an embedded system from scratch (PO11, PEO1, PEO2). Students will gain knowledge on embedded system functions, embedded components, embedded software, and coding which are required for practical implementation of a real embedded system (PO5, PO9, PSO2, PEO5). Students will acquire skills on embedded system coding, microcontrollers, sensors, and actuators to become a good embedded system engineer (PSO2, PEO3). Today, world leading companies are looking for the people who are expertise in the areas of embedded systems, embedded AI, and edge computing, etc (PO10, PO11). This course will provide a platform to develop skills for which companies are interested to hire professionals (PO6).		
Evaluation/ Grading Policy	Two in-semester examinations, a final examination, and lab assignments. Evaluation scheme: (i) First In-Sem Exam: 20%, (ii) Second In-Sem Exam: 20%, (iii) End-Sem Exam: 30%, and (iv) Labs: 30%.		
Course Materials/ References	Text Book: Frank Vahid and Tony Givargis, Embedded System Design: A Unified Hardware/Software Introduction, Wiley, Student edition, 2006. Reference Book: 1) Sarah Harris and David Harris, Digital Design and Computer Architecture: ARM Edition, Morgan Kaufmann Publishers Inc., United States, May 2015. 2) Jonathan Walker Valvano, Embedded Systems: Introduction to Arm® Cortex™-M Microcontrollers, 5th Edition, CreateSpace Independent Publishing Platform, 2011. 3) Samir Palnitkar, Digital Design using Verilog HDL, Prentice Hall; 2nd Edition, 2003. 4) ARM System Developer's Guide: Designing and Optimizing System Software 1st Edition (Designing and Optimizing System Software) Publisher: Morgan Kaufmann Publishers Inc., 2011.		
Detailed Course Content	APPENDIX		

Course Outcome:

The Course Outcomes (COs) are mapped to the course modules, identified as CO1, CO2, CO3, CO4, CO5, CO6, and CO7. The course weightage is distributed based on no. of lectures. The expected POs versus COs matrices are tabulated below.

POs-COs Matrix:

Mapping of Course Outcomes (COs) and Program Outcomes (POs)																	
Note: 1 = Slight 2 = Moderate 3 = STRONG																	
Course Outcomes	Modules	Hours	Program Outcomes (POs)												PSOs		
			PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	4	1	3	3	3	2	0	0	0	0	0	0	2	2	0	3
CO2	2	13	0	3	3	3	0	2	0	0	0	0	0	0	2	2	3
CO3	3	12	0	3	3	3	2	2	1	0	0	2	2	2	2	2	3
CO4	4	9	0	3	3	3	0	2	0	0	0	2	2	2	2	2	3
CO5	5	8	0	3	3	0	0	0	0	1	1	0	0	0	2	0	3
CO6	6	7	0	3	3	3	2	0	0	0	0	0	0	0	2	0	3
CO7	7	9	0	3	3	0	0	0	0	0	0	0	0	0	3	0	3
AVG			0.14	3.00	3.00	2.14	0.86	0.86	0.14	0.14	0.14	0.57	0.57	0.86	2.14	0.86	3.00

APPENDIX: Detailed Course Content

Sl. No	Description		No. of Lectures
Module-1	Introduction		4
	1.1	Embedded Systems Overview	1
	1.2	Design Challenges – Optimizing Design Metrics	1
	1.3	Processor and IC Technologies	1
	1.4	Design Technology and Trade-offs	1
Module-2	Custom Single-Purpose Processors		13
	2.1	Binary Information Implemented with MOS transistors	1
	2.2	Digital Logic, Logic gates	1
	2.3	Combinational and Sequential Logic	2
	2.4	RT-Level Combinational and Sequential Components	1
	2.5	Basic Computer Architecture	1
	2.6	Finite State Machines (FSM)	2
	2.7	Optimizing Datapath	2
	2.8	Hardware Description Languages: Verilog HDL	2
	2.9	Field Programmable Gate Array (FPGA)	1
Module-3	Microcontroller Based on Cortex-M Class Processors/AVR microcontroller		12
	3.1	AVR microcontroller, ARM/Cortex family microcontrollers	1
	3.2	AVR microcontroller, ARM Assembly Language	3
	3.3	Introduction to Embedded C	1
	3.4	Mixing C and Assembly	1
	3.5	Developing programs with ARM/Cortex microcontrollers	3
	3.6	Peripheral Drivers Development	2
	3.7	Introduction to Keil µVision	1
Module-4	Real-Time Interfacing		9
	4.1	Introduction to Input/Output	1
	4.2	SysTick Timer	1
	4.3	ARM interrupts and interrupt processing	1

	4.4	GPIO Polling and Interrupt	1
	4.5	UART-Serial Interface	1
	4.6	Analog I/O Interfacing, I/O Synchronization, RS232, I2C, CAN	2
	4.7	USB Protocols, Bluetooth	2
Module-5	Implementation of Embedded Systems		8
	5.1	Open-Loop and Closed-Loop Control Systems	1
	5.2	Example: Self-Driving Car	1
	5.3	ADC and DAC	1
	5.4	IMU (Inertial Measurement Unit), Image Sensor, Pressure Sensor	2
	5.5	Servo-Motor, Pulse-Width-Modulation (PWM)	1
	5.6	Proportional-Integral-Derivative (PID) Controller	1
	5.7	Integration of Sensors and Motors with RaspberryPi/Arduino Boards	1
Module-6	Memory		7
	6.1	Introduction	1
	6.2	Common Memory Types, ROM, RAM, EEPROM, SRAM	3
	6.3	Memory Hierarchy and Cache	1
	6.4	Advanced RAM, DRAM	1
	6.5	Memory Management	1
Module-7	IC Technology		9
	7.1	Full-Custom IC Technology	2
	7.2	Semi-Custom IC Technology	2
	7.3	Programmable Logic Device IC Technology	1
	7.4	Design Technology: Automation and Synthesis	2
	7.5	Verification: Hardware/Software Co-Simulation	2